QUANTUMFLOOR

Computational Research Report

Job ID: 073f4d5a

1. Executive Summary

This report details the execution of the QuantumFloor diagnostic pipeline on dataset 'target'. The system evaluated multiple classical baselines against quantum algorithms to determine the optimal strategy.

2. Dataset Intelligence

Modality:	TABULAR
Total Rows:	1025
Unique Rows:	302
Duplicate Rows:	723
Total Columns:	13
Missing Values:	0.00%
Leakage Risk Detected:	0

Scientific Integrity Warnings

Warning: Duplicate feature groups: 302 | Duplicate rows: 723 | Cross-split duplicate groups: 0 | Target leakage detected: 0 | Identifier leakage detected: 0 | Preprocessing leakage: 0

3. Data Preparation

Train/Test Split:	80/20 Stratified
Preprocessing:	Pipeline (Imputation + Scaling)
Leakage Protection:	Enforced

4. Computational Plan

Original Features:	13
Quantum Embedding Dimensions:	4
Qubit Budget:	4
Classical Training Size:	820
Quantum Subset Size:	120

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5. Quantum Execution

Execution Backend: SIMULATED EXECUTION

Ansatz: AngleEmbedding + IsingZZ

Qubits: 4

Circuit Depth: 4

Total Operations: 4

6. Classical vs Quantum Baselines

Model	Type	AUROC	F1	Spec@95	Notes
full_feature_classical	Classical	0.8883	0.8053	0.6211	
logreg	Classical	0.9168	0.8571	0.6526	
svm_rbf	Classical	0.9211	0.8531	0.6947	
random_forest	Classical	0.8635	0.8372	0.4526	
qsvm	Quantum	0.8128	0.7358	0.4526	
pqk	Quantum	0.8659	0.8037	0.3789	
vqc	Quantum	0.9189	0.8529	0.6526	

7. Comparison & Explainability

Best Classical AUROC: 0.9211

Best Quantum AUROC: 0.9189

Delta: -0.0023

Explainability Status: available

PC1: 0.2968

PC4: 0.0703

PC3: 0.0360

PC2: 0.0323

8. Final Verdict

Best Overall Model: svm_rbf

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Quantum Advantage: No